

Deterministic Optimization

Algebraic Aspects of Linear Optimization

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Basic Feasible Solution in
Standard Form LP

Algebraic Aspects of Linear Optimization

Learning Objectives for this lesson

- Find a BFS for a standard form LP

Example of a Standard Form LP

- A standard form LP:

- $\min x_1 + x_2 - x_3$
- S.t. $-x_1 + x_2 - x_3 + x_4 = 2$
- $-2x_1 + x_2 + x_3 - x_5 = 3$
- $x_1, x_2, x_3, x_4, x_5 \geq 0$

- $c = [1, 1, -1, 0, 0]^T$

- $A = \begin{bmatrix} -1 & 1 & -1 & 1 & 0 \\ -2 & 1 & 1 & 0 & -1 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Example cont.

- $A = \begin{bmatrix} -1 & 1 & -1 & 1 & 0 \\ -2 & 1 & 1 & 0 & -1 \end{bmatrix}$
- We can choose a basis matrix B as the first two columns of A
- $B = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$ corresponds to $x_B = [x_1, x_2]$
- Because the above B has 2 linearly independent columns, i.e. invertible
- $N = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ corresponds to $x_N = [x_3, x_4, x_5]$
- Then set $x_3 = 0, x_4 = 0, x_5 = 0$, these are 3 active constraints coming from the corresponding nonnegativity constraints.
- This set of 5 equations in 5 variables will give a basic solution.

Example cont.

- $A = \begin{bmatrix} -1 & 1 & -1 & 1 & 0 \\ -2 & 1 & 1 & 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$
- We can choose a basis matrix B as
- $B = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$ corresponds to $x_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
- $N = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ corresponds to $x_N = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Is This a Basic Feasible Solution?

- The basic solution we found is

$$\bullet \quad x = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Since $x_1 = -1 < 0$, this is NOT a feasible solution to the original LP
- Therefore, this basic solution is not a basic feasible solution
- We need to try another basis matrix and see if the resulting basic solution has all nonnegative components.

Try Another Basis Matrix

- Let us try another basis matrix
 - $B = [A_3 \ A_4] = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$
 - $B^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
 - $x_B = B^{-1}b = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$
 - $x_N = \begin{bmatrix} x_1 \\ x_2 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 - Because all components are nonnegative, this is a BFS.

Summary

- We learned:
 - In order to find a basic feasible solution, we can sample a basis and compute the corresponding basic solution. If the basic variables are all nonnegative, it is a BFS; otherwise it is a basic solution but not feasible. We need to try another basis matrix until a BFS is found.